

LECTURE 3: INDUCTION MOTOR DRIVES

Commutatorless motor drives offer a number of advantages over conventional DC commutator motor drives for the electric propulsions of EVs and HEVs. At present, induction motor drives are the mature technology among commutatorless motor drives. Compared with DC motor drives, the AC induction motor drive has additional advantages such as lightweight nature, small volume, low cost, and high efficiency. These advantages are particularly important for EV and HEV applications.

There are two types of induction motors, namely, wound-rotor and squirrel cage motors. Because of the high cost, need for maintenance, and lack of sturdiness, wound-rotor induction motors are less attractive than their squirrel-cage counterparts, especially for electric propulsion in EVs and HEVs. Hence, squirrel-cage induction motors are loosely named as induction motors.

A cross section of a two-pole induction motor is shown in Figure 1. Slots in the inner periphery of the stator are inserted with three phase windings, a-a', b-b', and c-c'. The turns of each winding are distributed such that the current in the winding produces an approximate sinusoidally distributed flux density around the periphery of the air gap. The three windings are spatially arranged by 120° as shown in Figure 1.

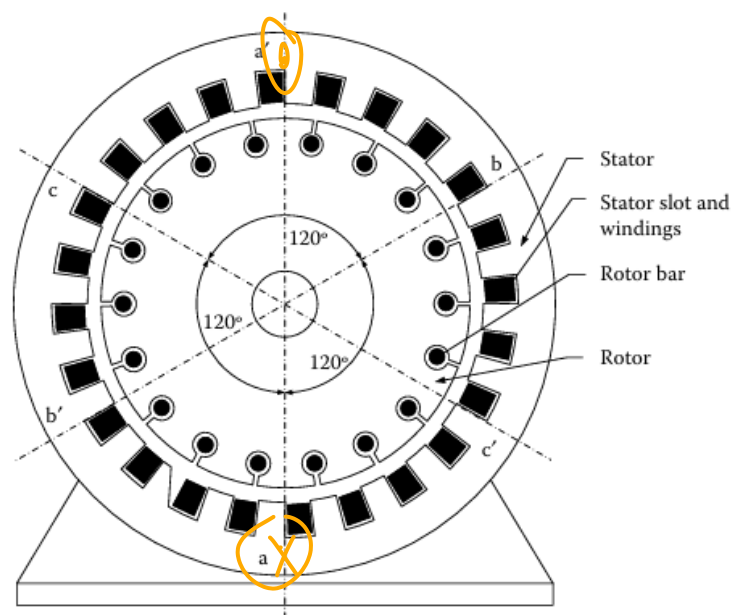


Figure 1: Cross section of an induction motor.

The most common types of induction motor rotors are the squirrel-cage motors in which aluminum bars are cast into slots in the outer periphery of the rotor. The aluminum bars are short-circuited together at both ends of the rotor by cast aluminum end rings, which also can be shaped as fans.

BASIC OPERATION PRINCIPLES OF INDUCTION MOTORS

Figure 2 shows, schematically, a cross section of the stator of a three phase, two-pole induction motor. Each phase is fed with a sinusoidal AC current, which has a frequency of ω and a 120° phase difference between each other. Current i_{as} , i_{bs} , and i_{cs} in the three stator coils a-a', b-b', and c-c' produce alternative magnetic motive forces (mmfs), F_{as} , F_{bs} , and F_{cs} , which are space vectors. The resultant stator mmf vector F_s^s constitutes a vector sum of the phase mmf vectors.

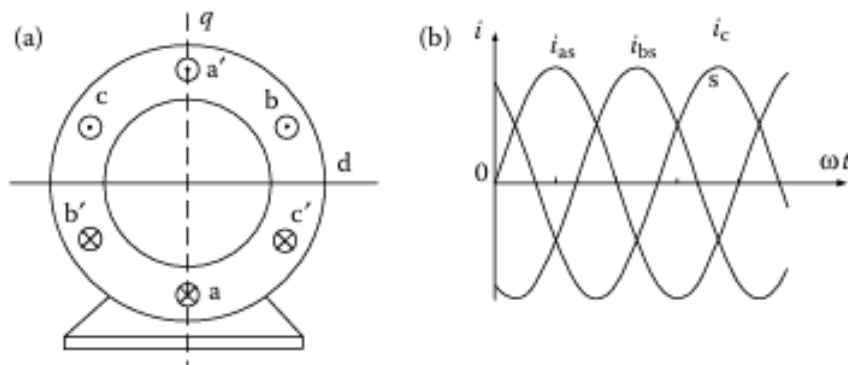


Figure 2: Induction motor stator and stator winding current:

(a) spatially symmetric three phase stator windings (b) phase currents

The mmfs produced by the phase currents can be written as

$$F_{as} = F_{as} \sin \omega t,$$

$$F_{bs} = F_{bs} \sin(\omega t - 120^\circ),$$

$$F_{cs} = F_{cs} \sin(\omega t - 240^\circ)$$

The resultant stator mmf vector, F_s^s , is expressed as

$$F_s^s = F_{as}^s e^{j0^\circ} + F_{bs}^s e^{j120^\circ} + F_{cs}^s e^{j240^\circ}$$

Assuming that the magnitudes of the three-phase mmfs are identical, equal to F_s , Equation can be further expressed as

$$F_s^s = \frac{3}{2} F_s e^{j(\omega t - 90^\circ)}$$

Equation indicates that the resultant stator mmf vector is rotating with the frequency of the angle velocity of ω , and its magnitude is $\frac{3}{2} F_s$.

Figure 3 graphically shows the stator mmf vectors at $\omega t = 0$ and $\omega t = 90^\circ$; here, ωt is the angle, rather than the resultant stator mmf vector relative to the d-axis. Actually, if the ωt is taken as the reference, the resultant stator mmf vector is a 90° delay to the phase a-a' mmf.

The reaction between the rotating stator mmf and the rotor conductors induces a voltage in the rotor and hence electric current in the rotor. In turn, the rotating mmf produces a torque on the rotor, which is carrying the induced current. It is clear that the induced current in the rotor is essential for producing torque, and in turn the induced current depends on the relative movements between the stator mmf and the rotor. That is why there must exist a difference between the angular velocity of the rotating stator mmf and the angular velocity of the rotor.

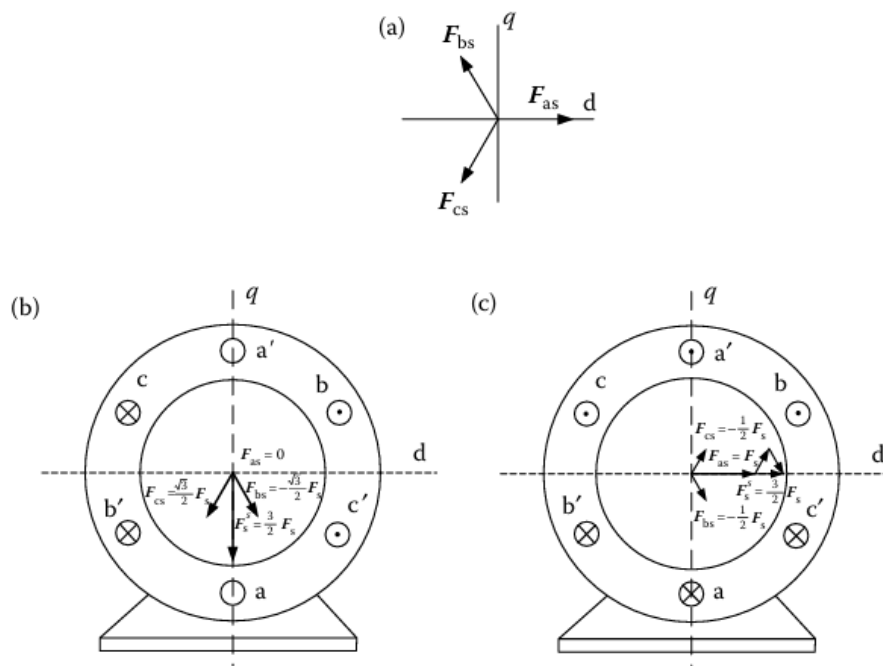


Figure 3: Stator mmf vectors: (a) positive direction of each phase mmf; (b) stator mmf vectors at $\omega t = 0$; (c) stator mmf vectors at $\omega t = 90^\circ$

The frequency ω , or angular velocity of the rotating stator mmf in equation, depends only on the frequency of the alternative current of the stator; thus, it is referred to as electrical angular velocity. For the machine with two poles, the electrical angular velocity is identical to the mechanical angular velocity of the rotating stator mmf. However, for the machine with more than two poles, the mechanical angular velocity differs from the electrical one, which can be expressed as

$$\theta_e = \frac{P}{2} \theta_{\text{mechanical}}$$

$$\omega_{ms} = \frac{2}{p} \omega = \frac{4\pi f}{p} \text{ rad/s}$$

where f is the frequency of the alternative current or angular velocity of the rotating stator mmf in *cycle/s*. When the angular velocity of the rotor is equal to the mechanical angular velocity of rotating stator mmf, there will be no induced current in the rotor, and then no torque is produced. Thus, the mechanical angular velocity of the rotating stator mmf is also called synchronous speed.

If the rotor speed is ω_m in *rad/s*, then the relative speed between the stator rotating field and the rotor is given by

$$\omega_{sl} = \omega_{ms} - \omega_m = s\omega_{ms}$$

where ω_{sl} is called slip speed.

The parameter s , known as slip, is given by

$$s = \frac{\omega_{ms} - \omega_m}{\omega_{ms}} = \frac{\omega_{sl}}{\omega_{ms}}$$

Because of the relative speed between the stator field and the rotor, balanced three-phase voltages are induced in the rotor mentioned before. The frequency of these voltages is proportional to the slip speed. Hence

$$\omega_r = \frac{\omega_{sl}}{\omega_{ms}} \omega = s\omega$$

where ω_r is the frequency of the rotor voltage induced.

For $\omega_m < \omega_{ms}$, the relative speed is positive; consequently, the rotor induced voltages have the same phase sequence as the stator voltages. The three-phase current flowing through the rotor produces a magnetic field, which moves with respect to the rotor at the slip speed in the

stator flux and rotor flux rotate at the exact same speed—synchronous speed (ω_{ms})—with respect to the stationary frame of the stator.

same direction as the rotor speed. Consequently, the rotor field moves in the space at the same speed as the stator, and a steady torque is produced.

For $\omega_m = \omega_{ms}$, the relative speed between the rotor and stator field becomes zero. Consequently, no voltages are induced and no torque is produced by the motor.

For $\omega_m > \omega_{ms}$, the relative speed between the stator field and the rotor speed reverses. Consequently, the rotor-induced voltages and currents also reverse and have a phase sequence opposite to that of the stator. Moreover, the developed torque has a negative sign, suggesting generator operation. (The generator is used to produce regenerative braking.)

STEADY-STATE PERFORMANCE

A per-phase equivalent circuit of an induction motor is shown in Figure 4a. The fields produced by the stator and rotor are linked together by an ideal transformer. a_{T1} is the transformer factor, which is equal to $\frac{n_s}{n_r}$, where n_s and n_r are the numbers of turns of stator and rotor windings, respectively. For a squirrel-cage rotor, $n_r = 1$. The equivalent circuit can be simplified by referring the rotor quantities to the stator frequency and number of turns.

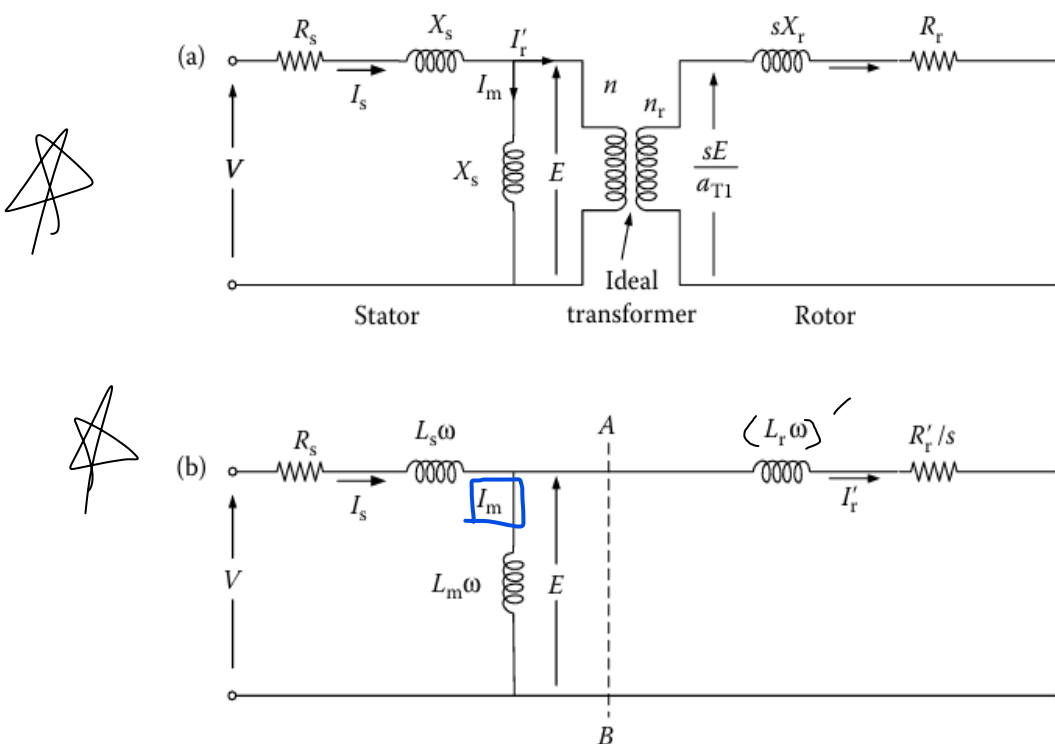


Figure 4: Per phase equivalent circuit of an induction motor: (a) with a transformer between rotor and stator and (b) refer rotor quantities to stator.

The resultant equivalent circuit is shown in Figure 4b, where R'_r and X'_r are the rotor resistance and reactance referred to the stator, and is given by the following equations:

$$R'_r = a_{T1}^2 R_r \text{ and } X'_r = a_{T1}^2 X_r$$

The stator reactance, mutual reactance, and rotor reactance referred to the stator can be expressed by the stator frequency and their inductances, L_s , L_m , and L_r , as shown in Figure 4.

The impedances of stator, field, and rotor can be expressed as

$$Z_s = R_s + jL_s\omega$$

$$Z_m = jL_m\omega$$

$$Z_r = \frac{R'_r}{s} + jL_r\omega$$

The driving-point impedance of the circuit is

$$Z = Z_s + \frac{Z_m Z_r}{Z_m + Z_r}$$

Hence, the current I_s and I'_r can be calculated as

$$I_s = \frac{V}{Z}$$

And

$$I'_r = \frac{Z_m}{Z_m + Z_r} I_s$$

The total electrical power supplied to the motor for three phases is

$$P_{elec} = 3I_r'^2 \frac{R'_r}{s}$$

The mechanical power of the rotor can be obtained by subtracting the total power loss in the stator as

$$P_{mech} = P_{elec} - 3I_r'^2 R'_r$$

The angular velocity of the rotor, ω_m , is

$$\omega_m = \frac{2}{p} \omega_s (1 - s)$$

The torque developed by the motor can be determined by

$$T = \frac{P_{mech}}{\omega_m}$$

Figure 5 shows the torque–slip characteristics of an induction motor, which has fixed voltage and frequency.

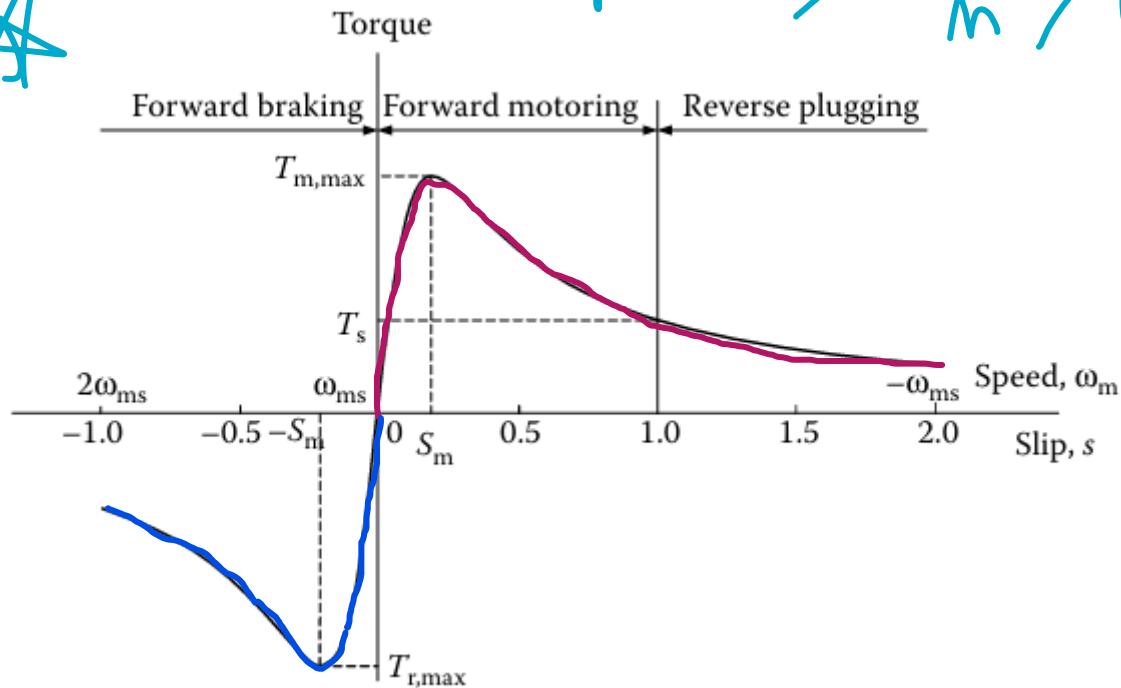


Figure 5: Torque–slip characteristics of an induction motor with fixed stator frequency and voltage

In the region of $0 < s < s_m$, where s_m is the rated slip of the motor, the torque increases approximately linearly with the increase of slip until reaching its maximum at $s = s_m$, then it decreases with the further increase of the slip. At $s = 1$, the rotor speed is zero and the corresponding torque is the starting torque, which is less than its torque at $s = s_m$. The region of $0 < s < 1$ is the forward motoring region.

In the region of $s > 1$, the rotor torque is positive and decreases further with the increase of slip, and the rotor speed is negative, according to equation $s = \frac{\omega_{ms} - \omega_m}{\omega_{ms}} = \frac{\omega_{sl}}{\omega_{ms}}$. Thus, in this region, the operation of the motor is reverse braking.

In the region of $s < 0$, that is, when the rotor speed is greater than the synchronous speed, the motor produces a negative torque.

$$r = s \times \max T_e$$

It is clear that the speed–torque characteristic of a fixed voltage and fixed frequency induction motor is not appropriate to vehicle traction applications. This is due to the low starting torque, limited speed range, and unstable operation in the range of $s > s_m$, in which any additional disturbing torque in the load will lead the machine to stop as the torque decreases with the speed decreasing characteristically.

The high slip also results in high current, which may cause damage in the stator windings. Actually, the operation of the fixed voltage and frequency induction motor are usually operated in the narrow slip range of $0 < s < s_m$. Thus, for traction application, an induction motor must be controlled to provide proper speed–torque characteristic.

CONSTANT VOLT/HERTZ CONTROL

For traction application, the torque–speed characteristic of an induction motor can be varied by simultaneously controlling the voltage and frequency, which is known as the constant volt/hertz control. By emulating a DC motor at low speed, the flux may be kept constant. According to Figure 4b, the field current I_m should be kept constant and equal to its rated value. That is,

$$I_{mr} = \frac{E}{X_m} = \frac{E_{rated}}{\omega_r L_m}$$

where I_{mr} is the rated field current, E_{rated} and ω_r are the rated mmf and frequency of the stator, respectively.

To maintain the flux at constant, the $\frac{E}{\omega}$ should be kept constant and equal to $\frac{E_{rated}}{\omega_r}$. Ignoring the voltage drops in the stator impedance Z_s results in a constant $\frac{V}{\omega}$ until the frequency and voltage reach their rated values. This approach is known as constant volt/hertz control.

From Figure 4b, the rotor current can be calculated as

$$I'_r = \frac{\left(\frac{\omega}{\omega_r}\right) E_{rated}}{jL_r\omega + \frac{R'_r}{s}}$$

The torque produced can be obtained as

$$T = \frac{3}{\omega} I_r'^2 \frac{R_r'}{s} = \frac{3}{\omega} \left[\frac{\frac{E_{rated}^2 R_r'}{s}}{\left(\frac{R_r'}{s}\right)^2 + (L_r' \omega)^2} \right]$$

The slip s_m corresponding to the maximum torque is

$$R_r' = s_m X_r'$$

And then, the maximum torque is

$$T_{max} = \frac{3}{2} \frac{E_{rated}^2}{L_r' \omega_r^2}$$

Equation indicates that with the constant $\frac{E}{\omega}$, the maximum torque is constant with varying frequency. Equation s_m indicates that $s_m \omega$ is constant, resulting in constant slip speed, ω_{sl} . In practice, due to the presence of stator impedance and the voltage drop, the voltage should be somewhat higher than that determined by constant E/ω , as shown in Figure 6.

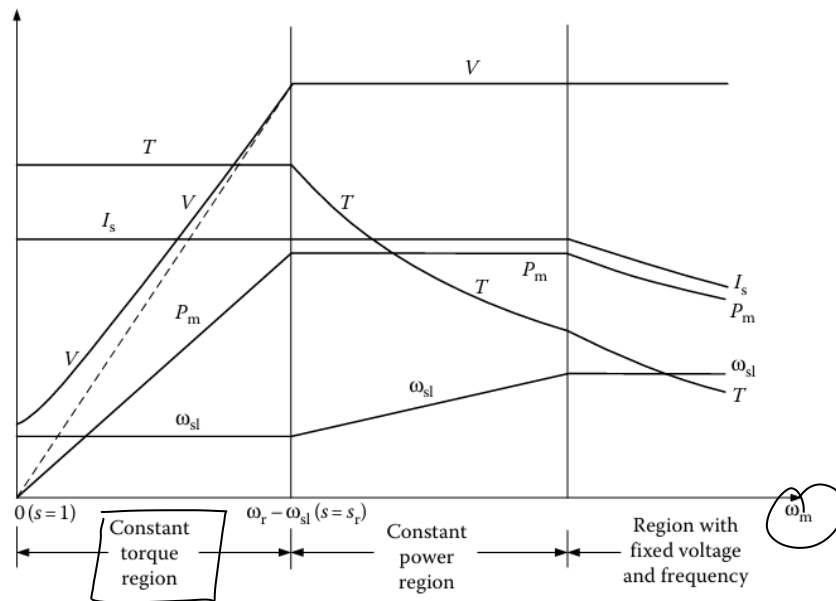


Figure 6: Operating vehicle varying with motor speed

When motor speed is beyond its rated speed, the voltage reaches its rated value and cannot be increased with the frequency. In this case, the voltage is fixed to its rated value and the frequency increases continuously with the motor speed. The motor goes into the field weakening operation. The slip s is fixed to its rated value corresponding to the rated frequency,

and the slip speed ω_{sl} increases linearly with the motor speed. This control approach results in constant power operation as shown in Figure 6.

In traction applications, speed control in a wide range is usually required and the torque demand in the high-speed range is low. Control beyond constant power range is required. To prevent the torque from exceeding the breakdown torque, the machine is operated at a constant slip speed and the machine current and power are allowed to decrease as shown in Figure 6. Figure 7 shows a general block diagram where constant V/f control is implemented.

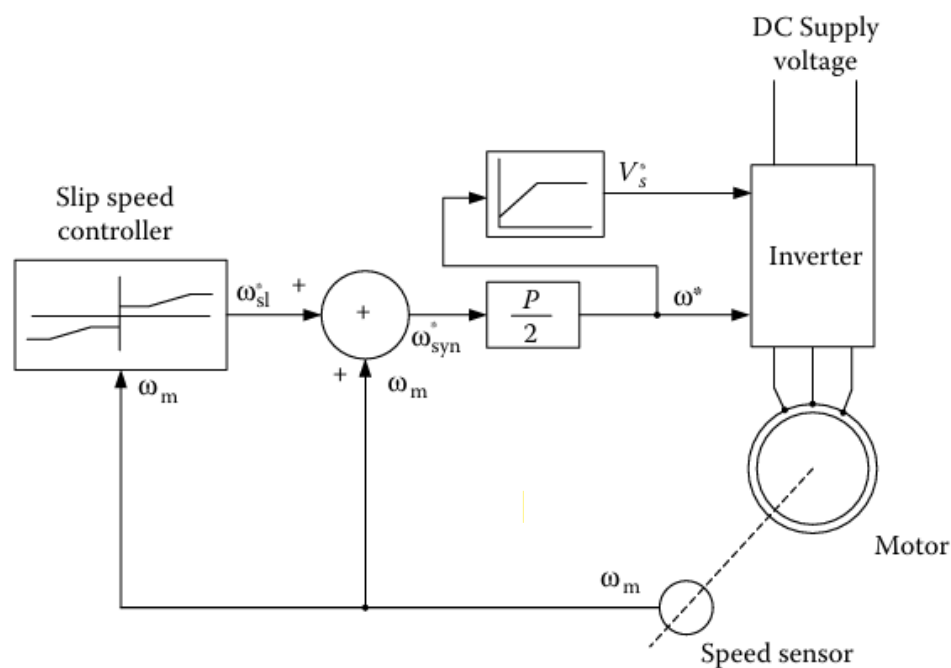


Figure 7: General configuration of constant V/f control.

POWER ELECTRONIC CONTROL

As EV and HEV propulsion, an induction motor drive is usually fed with a DC source (battery, fuelcell, etc.), which has approximately constant terminal voltage. Thus, a variable frequency and variable voltage DC/AC inverter is needed to feed the induction motor. The general DC/AC inverter is constituted by power electronic switches and power diodes. The commonly used topology of a DC/AC inverter is shown in Figure 8a, which has three legs (S_1 and S_4 , S_3 and S_6 , and S_5 and S_2), feeding the phase a, phase b, and phase c of the induction motor. When switches S_1 , S_3 , and S_5 are closed, S_4 , S_6 , and S_2 are opened, and phases a, b, and c are supplied with a positive voltage ($V_d/2$). Similarly, when S_1 , S_3 , and S_5 are opened and S_4 , S_6 , and S_2 are

closed, the phases a, b, and c is supplied with a negative voltage. All the diodes provide a path for the reverse current of each phase.

For constant volt/hertz control of an induction motor, sinusoidal pulse width modulation (PWM) is used exclusively. Three-phase reference voltage V_a, V_b and V_c of variable amplitude A_a, A_b , and A_c are compared with a common isosceles triangular carrier wave V_{tr} of a fixed amplitude A_m as shown in Figure 8c. The outputs of comparators 1,2, and 3 form the control signals for the three legs of the inverter. When the sinusoidal reference voltage V_a, V_b , and V_c , at a time t is greater than the triangular waved voltage, turn-on signals are sent to the switches S_1, S_3 , and S_5 and turn-off signals to S_4, S_6 , and S_2 . Thus, the three phases of the induction motor have positive voltage. On the other hand, when the reference sinusoidal voltage is smaller than the triangular wave voltage, turn-on signals are sent to the switches S_1, S_3 , and S_5 and turn-off signals to S_4, S_6 , and S_2 . The three phases of the induction motor then have a negative voltage. The voltages of the three phases are shown in Figure 8d through 8f.

The frequency of the fundamental component of the motor terminal voltage is the same as that of the reference sinusoidal voltage. Hence, the frequency of the motor voltage can be changed by changing the frequency of the reference voltage. The ratio of the amplitude of the reference wave to that of the triangular carrier wave, m , is called the modulation index; therefore,

$$m = \frac{A}{A_m}$$

where A is the multitude of the reference sinusoidal voltage, V_a, V_b , or V_c , and A_m is the multitude of angular carrier voltage. The fundamental (rms) component in the phase waveform V_{ao}, V_{bo} , or V_{co} is given by

$$V_f = \frac{mV_d}{2\sqrt{2}}$$

Thus, the fundamental voltage increases linearly with m until $m = 1$ (i.e., when the amplitude of the reference wave becomes equal to that of the carrier wave). For $m > 1$, the number of pulses in V_{ao}, V_{bo} , or V_{co} , becomes less and the modulation ceases to be sinusoidal.

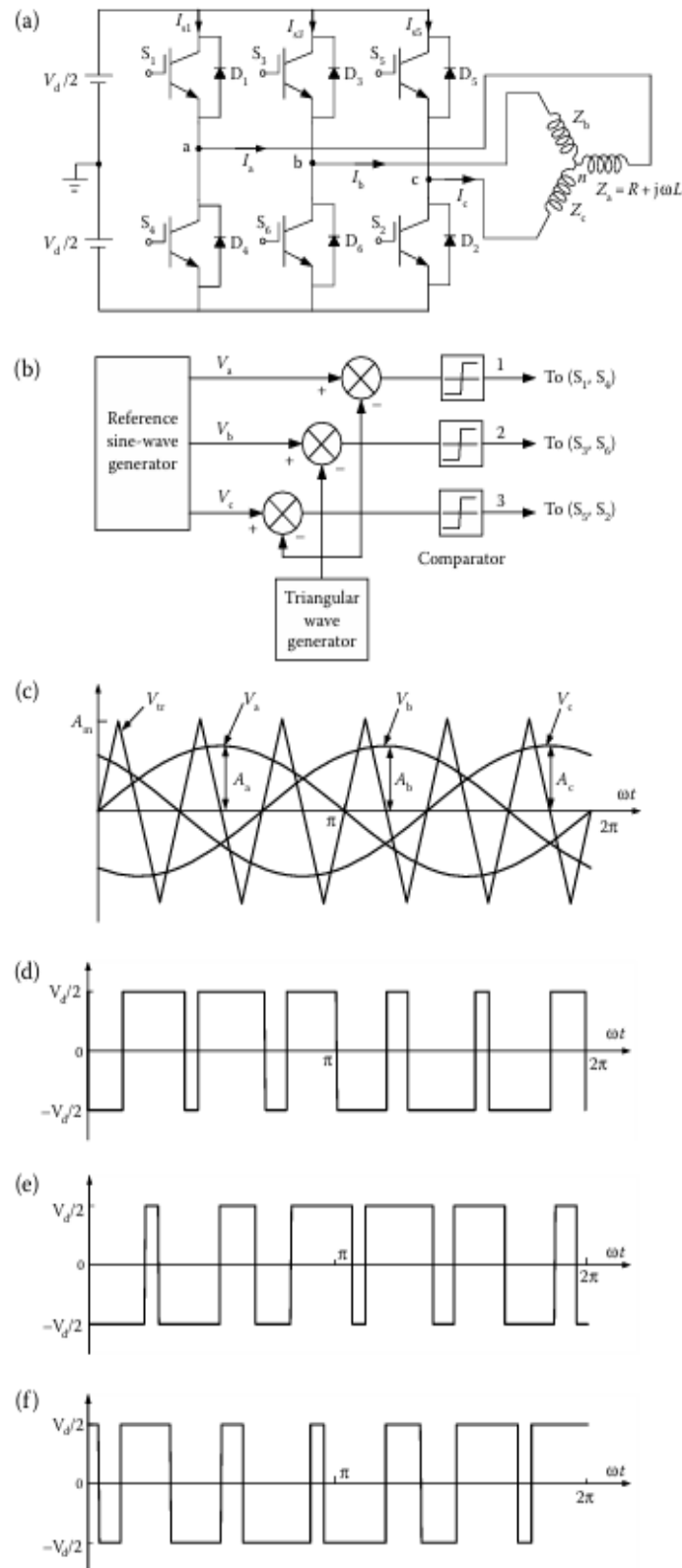


Figure 8 DC/AC inverter with sinusoidal pulse-width modulation: (a) inverter topology; (b) control signals; (c) three phase reference voltage and triangular carrier waveforms; (d) voltage of phase a; (e) voltage of phase b; and (f) voltage of phase c.

FIELD ORIENTATION CONTROL

The constant volt/hertz control of the induction motor is more suitably applied to the motors that operate with relative slow speed regulation. However, this approach shows poor response to frequent and fast speed varying, and also results in poor operation efficiency due to the poor power factor. In the last two decades, FOC or vector control technology has been successfully developed. This technology mostly overcomes the disadvantages of the constant volt/hertz control in AC motor drives.

FIELD ORIENTATION PRINCIPLES

The aim of FOC is to maintain the stator field perpendicular to the rotor field so as to always produce the maximum torque as in DC motors. However, for induction motors, phase voltages are the only accesses for the purpose of control.

When balanced three-phase sinusoidal currents flow through the three phases of the stator of an induction motor, the rotating field is developed; current is induced in the rotor. In turn, the current induced in the rotor is also three-phase and produces a field, which rotates with the same angular velocity of the stator rotating field. The rotating fields of both stator and rotor can be described by two rotating vectors, referring to a common, stationary reference frame, d-q, as shown in Figure 3.

The mmf of the stator field is expressed:

$$\mathbf{F}_s^s = \mathbf{F}_{as}e^{j0^\circ} + \mathbf{F}_{bs}e^{j120^\circ} + \mathbf{F}_{cs}e^{j240^\circ}$$

Similarly, the stator voltage, stator current, and stator flux can be expressed as vectors in the same way. That is,

$$\mathbf{v}_s^s = v_{as}^s e^{j0^\circ} + v_{bs}^s e^{j120^\circ} + v_{cs}^s e^{j240^\circ}$$

$$\mathbf{i}_s^s = i_{as}^s e^{j0^\circ} + i_{bs}^s e^{j120^\circ} + i_{cs}^s e^{j240^\circ}$$

$$\boldsymbol{\lambda}_s^s = \lambda_{as}^s e^{j0^\circ} + \lambda_{bs}^s e^{j120^\circ} + \lambda_{cs}^s e^{j240^\circ}$$

The subscript s refers to the stator and a_s , b_s , and c_s refer to phases a, b, and c of the stator. The superscript s refers to the variable that is referred to the stator fixed frame. Bold symbols stand

for vector variables. The vectors of stator voltage, current, and flux can be also described by its components in d and q axes as follows:

$$\begin{bmatrix} v_{ds}^s \\ v_{qs}^s \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} v_{as} \\ v_{bs} \\ v_{cs} \end{bmatrix}$$

$$\begin{bmatrix} i_{ds}^s \\ i_{qs}^s \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_{as} \\ i_{bs} \\ i_{cs} \end{bmatrix}$$

$$\begin{bmatrix} \lambda_{ds}^s \\ \lambda_{qs}^s \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} \lambda_{as} \\ \lambda_{bs} \\ \lambda_{cs} \end{bmatrix}$$

In a real induction motor, the rotor winding differs from the stator winding, that is, the effective number of turns per phase of the rotor winding, N_r , is not equal to that of the stator winding, N_s . Therefore, the turns ratio, $v = N_s/N_r$, must be taken into account. The vectors of rotor current, voltage, and magnetic flux can be described by i_r^r , v_r^r , and λ_r^r in the rotor frame.

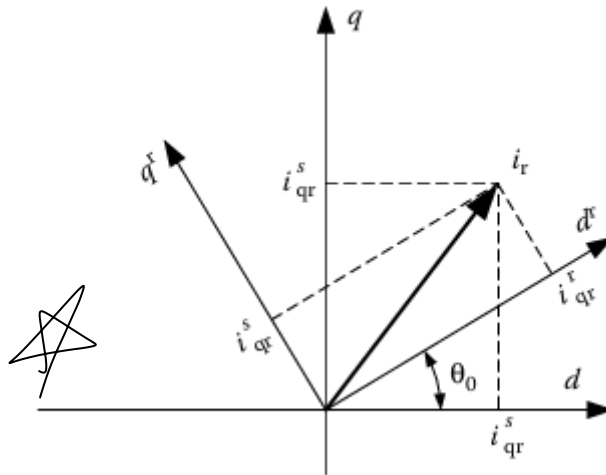


Figure 9: Transformation of rotor current vector from rotor frame to stator frame.

However, it is necessary to transform the vectors from the rotor frame to the stator frame for the purpose of easy analysis. The transformations of these vectors (refer to Figure 9) are described by

$$\mathbf{i}_r^s = \frac{e^{j\theta_0}}{v} \mathbf{i}_r^r,$$

$$\mathbf{v}_r^s = v e^{j\theta_0} \mathbf{v}_r^r$$

$$\lambda_r^s = v e^{j\theta_0} \lambda_r^r$$

Using vector notation, either the stator or rotor windings can be represented by a simple resistive-plus-inductive circuit, using current, voltage, and magnetic flux space vector as illustrated in Figure 10.

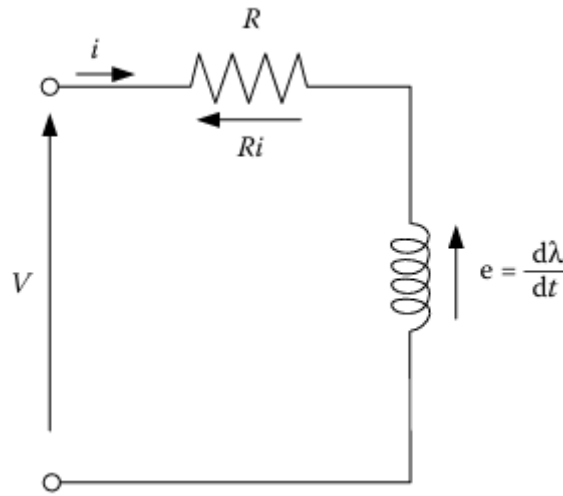


Figure 10: Resistive-plus-inductive equivalent circuit of either the stator or rotor windings.

Using the vector version of Kirchhoff's voltage law, the equation of the stator winding can be written as

$$\mathbf{v}_s^s = R_s \mathbf{i}_s^s + \frac{d\lambda_s^s}{dt}$$

and that of rotor winding as

$$\mathbf{v}_r^r = R_r^r \mathbf{i}_r^r + \frac{d\lambda_r^r}{dt}$$

where R_s and R_r^r are the actual stator and rotor resistances per phase, respectively. As is known from the steady-state theory of induction machines, the relation between R_r^r and the rotor resistance referred to the stator is

$$R_r^r = \frac{1}{v^2} R_r$$

Hence from Equations $\mathbf{i}_r^s = \frac{e^{j\theta_0}}{v} \mathbf{i}_r^r$, and $R_r^r = \frac{1}{v^2} R_r$,

$$R_r^r \mathbf{i}_r^r = \frac{e^{-j\theta_0}}{v} R_r \mathbf{i}_r^s$$

The second term, from Equation $\lambda_r^s = v e^{j\theta_0} \lambda_r^r$, gives

$$\frac{d\lambda_r^r}{dt} = \frac{e^{j\theta_0}}{v} \left(\frac{d\lambda_r^s}{dt} - j\omega_0 \right)$$

Finally, substituting above two equations into Equation $\mathbf{v}_r^r = R_r^r \mathbf{i}_r^r + \frac{d\lambda_r^r}{dt}$ gives

$$\mathbf{v}_r^s = R_r \mathbf{i}_r^s + \frac{d\lambda_r^s}{dt} - j\omega_0 \lambda_r^s$$

Introducing a differentiation operator $p \equiv d/dt$, the voltage equation of an induction motor can be written as

$$\mathbf{v}_s^s = R_s \mathbf{i}_s^s + p \lambda_s^s$$

$$\mathbf{v}_r^s = R_r \mathbf{i}_r^s + (p - j\omega_0) \lambda_r^s$$

The flux vector λ_s^s and λ_r^s can then be expressed in terms of current vector $\mathbf{i}_s^s$ and $\mathbf{i}_r^s$, and the motor inductances as

$$\begin{bmatrix} \lambda_s^s \\ \lambda_r^s \end{bmatrix} = \begin{bmatrix} L_s & L_m \\ L_m & L_r \end{bmatrix} \begin{bmatrix} \mathbf{i}_s^s \\ \mathbf{i}_r^s \end{bmatrix}$$

Where L_m is the mutual inductance; L_s is the stator inductance calculated as the sum of the stator leakage inductance L_{ls} and the mutual inductance L_m ; and L_r is the rotor inductance calculated as the sum of the rotor leakage inductance L_{lr} and the mutual inductance L_m .

Finally, the voltage equation can be written in matrix format as

$$\begin{bmatrix} v_{ds}^s \\ v_{qs}^s \\ v_{dr}^s \\ v_{qr}^s \end{bmatrix} = \begin{bmatrix} R_s & 0 & 0 & 0 \\ 0 & R_s & 0 & 0 \\ 0 & \omega_0 L_m & R_r & \omega_0 L_r \\ -\omega_0 L_m & 0 & -\omega_0 L_r & R_r \end{bmatrix} \begin{bmatrix} i_{ds}^s \\ i_{qs}^s \\ i_{dr}^s \\ i_{qr}^s \end{bmatrix} + \begin{bmatrix} L_s & 0 & L_m & 0 \\ 0 & L_s & 0 & L_m \\ L_m & 0 & L_r & 0 \\ 0 & L_m & 0 & L_r \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_{ds}^s \\ i_{qs}^s \\ i_{dr}^s \\ i_{qr}^s \end{bmatrix}$$

Because the rotor circuit of the induction motor is shorted, v_{dr}^s and v_{qr}^s are zero. At a given rotor speed, ω_0 , the stator and rotor currents can be obtained by solving above equation. The torque developed by the motor can be expressed as

$$T = \frac{2P}{\omega_s} L_m (i_{qs}^s i_{dr}^s - i_{ds}^s i_{qr}^s) = \frac{2P}{\omega_s} L_m I_m (i_s^s i_r^{s*})$$

where I_m stands for the imaginary part of the production of vector i_s^s and conjugate vector of i_r^{s*} .

Transferring three-phase variables (voltage, current, and flux) into a stationary stator-based dq frame does not change the alternate characteristics of the variable with time. AC quantities are somewhat inconvenient for control purposes. For instance, control systems are usually represented by block diagrams in which the variables are time-varying DC signals. Therefore, another transformation is necessary, which allows the conversion of the AC dq components of the motor vectors into DC variables. To do this, a transformation is conducted from a stationary stator reference frame, dq , to the so-called excitation reference frame DQ, which rotates with the angular speed ω , in the same direction as does mmf, F_s^s . As a result, in the steady state, coordinates of motor vectors in the new reference frame do not vary in time. This is illustrated in Figure 11, which shows the stator mmf vector in both reference frames.

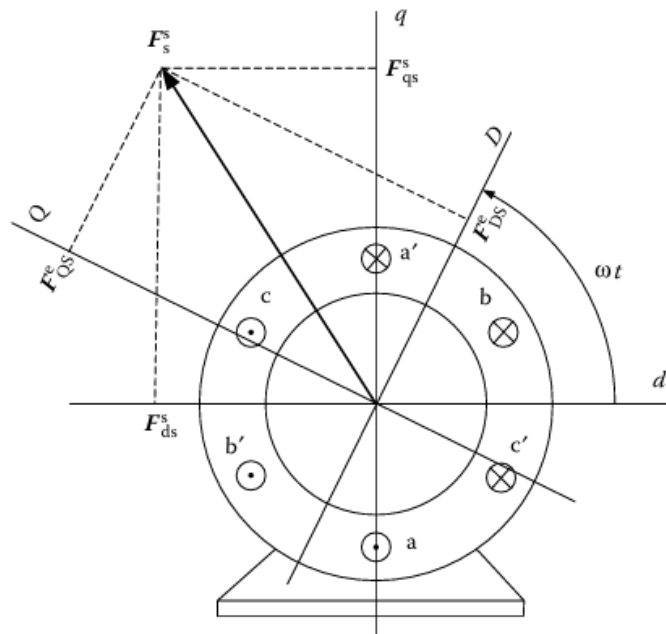


Figure 11: Stator mmf vector in the stator and excitation reference frames.

The voltage vector of the stator in excitation reference frame can be expressed as

$$v_s^e = v_s^s e^{-j\omega t}$$

Considering $e^{-j\omega t} = \cos(\omega t) - j\sin(\omega t)$, the components of stator voltage on DQ frame is

$$\begin{bmatrix} v_{ds}^e \\ v_{qs}^e \end{bmatrix} = \begin{bmatrix} \cos(\omega t) & \sin(\omega t) \\ -\sin(\omega t) & \cos(\omega t) \end{bmatrix} \begin{bmatrix} v_{ds}^s \\ v_{qs}^s \end{bmatrix}$$

Thus, the motor equation in excitation reference frame can be expressed as

$$v_s^e = R_s i_s^e + (p + j\omega) \lambda_s^e,$$

$$v_R^e = R_r i_R^e + (p + j\omega - j\omega_0) \lambda_R^e = R_r i_R^e + (p + j\omega_r) \lambda_R^e$$

where $\omega_r = \omega - \omega_0$ as the slip speed, and

$$\begin{bmatrix} \lambda_s^e \\ \lambda_R^e \end{bmatrix} = \begin{bmatrix} L_s & L_m \\ L_m & L_r \end{bmatrix} \begin{bmatrix} i_s^e \\ i_R^e \end{bmatrix}$$

Again, the rotor voltage vector is normally assumed to be zero because of the shorted rotor winding.

The torque equation in the excitation reference frame is similar to that in the stator frame as

$$T = \frac{P}{3} L_m (i_{qs}^e i_{DR}^e - i_{ds}^e i_{qR}^e) = \frac{P}{3} L_m I_m (i_s^e i_e^{e*})$$

In general, accurate control of instantaneous torque produced by a motor is required in high-performance drive systems, such as EV and HEV propulsions. The torque developed in the motor is a result of the interaction between current in the armature winding and the magnetic field produced in the stator field of the motor. The field should be maintained at a certain optimal level, sufficiently high to yield a high torque per unit of ampere, but not too high to result in excessive saturation of the magnetic circuit of the motor. With a fixed field, the torque is proportional to the armature current.

Independent control of the field and armature currents is desirable. In a similar manner to that of a DC motor, the armature winding in induction motors is also on the rotor, while the field is generated by currents in the stator winding. However, the rotor current is not directly derived from an external source, but results from the EMF induced in the winding as a result of the relative motion of the rotor conductors with respect to the stator field. In the most commonly

used squirrel-cage motors, only the stator current can be directly controlled, since the rotor winding is not accessible. Optimal torque production conditions are not inherent due to the absence of a fixed physical disposition between the stator and rotor fields, and the torque equation is nonlinear. FOC or vector control can realize the optimal control for transient operation of an induction drive. The FOC can decouple the field control from the torque control. A field-oriented induction motor emulates a separately excited DC motor in two aspects:

1. Both the magnetic field and the torque developed in the motor can be controlled independently.
2. Optimal conditions for torque production, resulting in the maximum torque per unit ampere, occur in the motor both in the steady state and in transient conditions of operations.

The optimal torque production conditions are inherently satisfied in a DC. The armature current i_a , supplied through brushes, is always orthogonal to the flux vector (field flux), λ_f , produced in the stator and linking the rotor winding. In effect, the developed torque, T , is proportional both to the armature current and the field flux, that is,

$$T = K_T i_a \lambda_f,$$

where K_T is a constant depending on the physical parameters of the motor. Thus, the torque of separately excited DC motors can be controlled by independently controlling the armature current and flux.

To emulate this independent armature and field control characteristic of a DC motor, the torque equation can be rearranged so that the torque is expressed in terms of the stator current and rotor flux. The following equation can be obtained as

$$i_R^e = \frac{1}{L_r} (\lambda_R^e - L_m i_S^e).$$

Torque Equation can be rewritten as

$$T = \frac{P}{3R_r} \frac{L_m}{\tau_r} (i_{QS}^e \lambda_{DR}^e - i_{DS}^e \lambda_{QR}^e)$$

where $\tau_r = L_r/R_r$ is the rotor time constant.

In Torque Equation, if $\lambda_{QR}^e = 0$, then

$$T = \frac{P}{3R_r} \frac{L_m}{\tau_r} (i_{QS}^e \lambda_{DR}^e)$$

Clearly, Equation is analogous to $T = K_T i_a \lambda_f$, describing a separately excited DC motor.

Substituting $v_R^e = 0$ (shorted rotor winding) into $v_R^e = R_r i_R^e + (p + j\omega_r) \lambda_R^e$ yields

$$R_r i_R^e + (p + j\omega_r) \lambda_R^e = 0$$

and substituting $i_R^e = \frac{1}{L_r} (\lambda_R^e R - L_m i_S^e)$ into Equation yields

$$p \lambda_R^e = \frac{1}{\tau_r} [L_m i_S^e - (1 + j\omega_r) \tau_r \lambda_R^e]$$

Thus,

$$p \lambda_{DR}^e = \frac{L_m}{\tau_r} i_{DS}^e - \frac{1}{\tau_r} \lambda_{DR}^e$$

Equation indicates that the flux λ_{DR}^e is produced by the current, i_{DS}^e . Thus, the torque produced can be represented by the block diagram as shown in Figure 12.

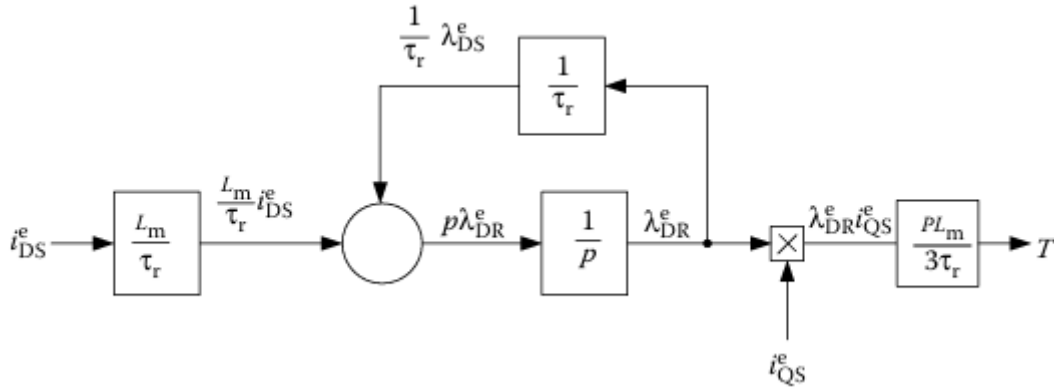


Figure 12: Block diagram of an induction motor with $\lambda_{QR}^e = 0$

Furthermore, Equation can be expressed into a transfer function as

$$G(p) = \frac{\lambda_{DR}^e}{i_{DS}^e} = \frac{L_m}{\tau_r p + 1}$$

Thus, the block diagram in Figure 12 can be further reduced as shown in Figure 13.

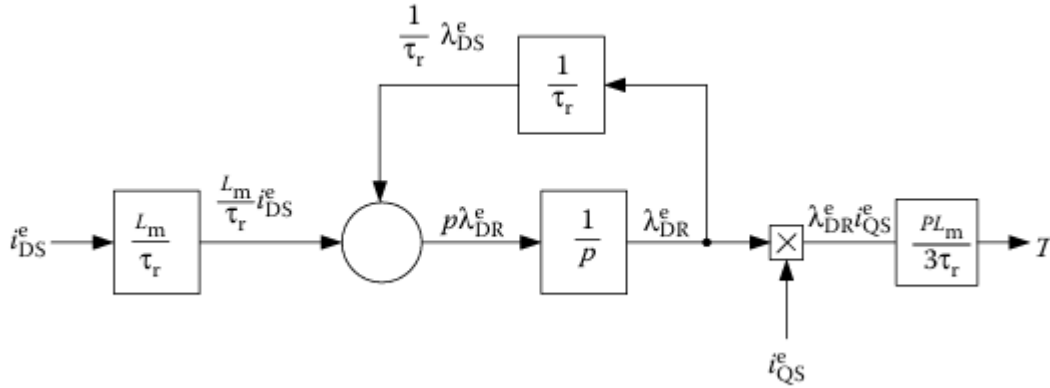


Figure 13: Block diagram of a field-orientated induction motor.

If conditions $\lambda_{QR}^e = 0$ and $\lambda_{DR}^e = \text{constant } t$ are satisfied, that is, $\lambda_{QR}^e = 0$ and $p\lambda_{DR}^e = 0$, then Equation $v_s^e = R_s i_s^e + (p + j\omega)\lambda_s^e$, yields $i_{DR}^e = 0$, that is, $i_R^e = j i_{QR}^e$. At the same time, $\lambda_R^e = \lambda_{DR}^e$. Consequently, vectors i_R^e and λ_R^e are orthogonal, which represents the optimal conditions for the torque production, analogous to a DC motor. In an induction motor, the optimal torque-production conditions are always satisfied in the steady state. However, in transient operation, the motor needs delicate control to achieve this optimal torque production.

CONTROL

As demonstrated in the above section, the field orientation principle defines the conditions of optimal torque production. Orthogonality of the rotor current and stator flux vectors must be maintained at all times. This is inherently satisfied in the steady state when the rotor settles down to such a speed that the developed torque matches the load torque. Under transient conditions, however, in order to meet the field orientation principle conditions, special techniques are required to provide an algorithmic equivalent of the actual physical disposition between the stator and rotor fields of the emulated DC motor.

The general block diagram of a vector control system for an induction motor drive is shown in Figure 14. A field orientation system produces reference signals, i_{as}^* , i_{bs}^* , and i_{cs}^* , of the stator currents, based on the input reference values, λ_r^* and T^* , of the rotor flux and motor torque, respectively, and the signals corresponding to selected variables of the motor. An inverter supplies the motor currents, i_{as} , i_{bs} , and i_{cs} , such that their waveforms follow the reference waveform, i_{as}^* , i_{bs}^* , and i_{cs}^* .

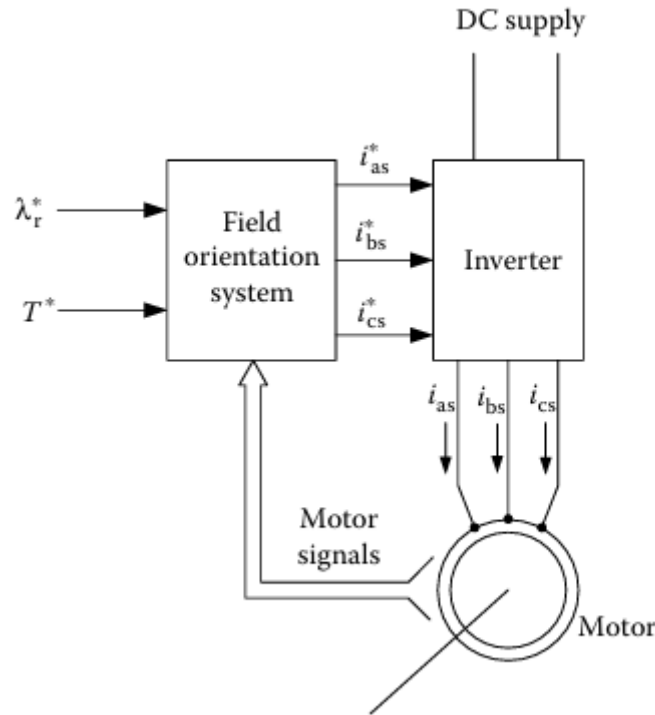


Figure 14: General block diagram of a vector control system for an induction motor.

As shown in Figure 13, in a field orientated induction motor, the i_{DS}^e and i_{QS}^e components of the stator current vector, i_S^e , in the excitation frame can be used for independent control of the motor field and torque, respectively. Hence, the field orientation system as shown in Figure 14 first converts λ_r^* and T^* into the corresponding reference signals, i_{DS}^{e*} and i_{QS}^{e*} , of the vector of the stator current and then transfers these into the reference signals, i_{as}^* , i_{bs}^* , and i_{cs}^* , of the stator phase current, which are to be produced by the inverter. The stator phase currents, i_{as}^* , i_{bs}^* , and i_{cs}^* , can be calculated using dq to abc transformation if the corresponding reference signals, i_{DS}^{e*} and i_{QS}^{e*} , in the stator reference frame are known. This is a simple scalar, or static, transformation, since the elements of the transformation matrix used to perform this operation are constant.

However, it can be seen that a dynamic transformation, that is, one involving time, is required to determine i_{DS}^{e*} and i_{QS}^{e*} from i_{DS}^{e*} and i_{QS}^{e*} .

In Figure 11, it does not indicate which vector the excitation reference frame DQ aligns with. Clearly, anyone of the vectors can be used as a reference with which the excitation frame is to

be aligned. Usually, it is the rotor flux vector, λ_r^s , along which the excitation frame is orientated. This method is usually referred to as the rotor flux orientation as shown in Figure 6.28.

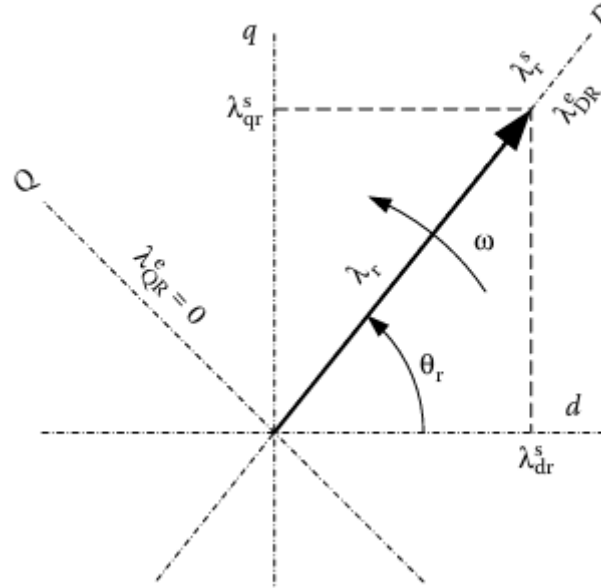


Figure 15: Orientation of the excitation reference frame along the rotor flux vector

If the angular position of the rotor flux vector in the stator reference frame is denoted by θ_r , the DQ to dq transformation in the described scheme is

expressed as

$$\begin{bmatrix} i_{ds}^{s*} \\ i_{qs}^{s*} \end{bmatrix} = \begin{bmatrix} \cos(\theta_r) & -\sin(\theta_r) \\ \sin(\theta_r) & \cos(\theta_r) \end{bmatrix} \begin{bmatrix} i_{DS}^{e*} \\ i_{QS}^{e*} \end{bmatrix}$$

It can be observed that this orientation of the orientation frame inherently satisfies the field orientation principle condition $\lambda_{QR}^e = 0$. The rotor flux is controlled by adjusting the i_{DS}^e component of the stator current vector— independently of the torque control, which is realized by means of the i_{QS}^e component. The only requirement for this scheme is an accurate identification of angle θ_r , that is, the position of λ_r^s . This can be done in either a direct or indirect way.

DIRECTION ROTOR FLUX ORIENTATION SCHEME

In direct field orientation systems, the magnitude and angular position (phase) of the reference flux vector, λ_r^e , are either measured or estimated from the stator voltage and current using flux

observers. For example, Hall sensors can be used to measure magnetic fields. Placing the sensors in the air gap of the motor, on the d and q axes, allows determination of the corresponding components of vector λ_m^s of the mutual flux (air gap flux). However, this air gap flux differs from the rotor flux, which is taken as the reference flux vector and needs derivation from the air gap flux λ_m^s .

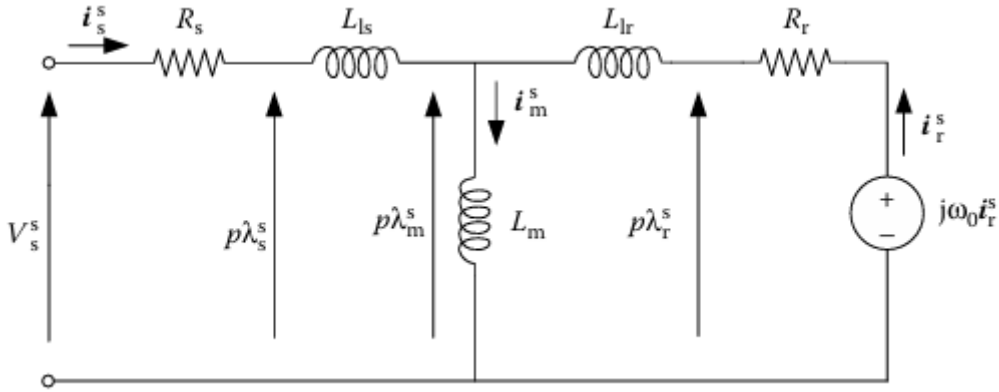


Figure 16: Dynamic T equivalent circuit of an induction motor.

Referring to the dynamic T equivalent circuit shown in Figure 16, the flux appearing across the mutual inductance L_m is

$$\lambda_m^s = L_m i_m^s = L_m (i_s^s + i_r^s)$$

or

$$i_r^s = \frac{1}{L_m} \lambda_m^s - i_s^s$$

Since λ_r^s differs from the λ_m^s by only the leakage flux in the rotor, then

$$\lambda_r^s = \lambda_m^s + L_{lr} i_r^s = \lambda_m^s + L_{lr} \left(\frac{1}{L_m} \lambda_m^s - i_s^s \right) = \frac{L_r}{L_m} \lambda_m^s - L_{lr} i_s^s$$

A microprocessor-based rotor flux calculator is shown in Figure 17.

It performs the algebraic operations as the follows:

1. Signals i_{ds}^s and i_{qs}^s are calculated from the actual stator phase currents, i_{as} , i_{bs} , and i_{cs} , using the abc to dq transformation.
2. Using above equation, signals λ_{dr}^s and λ_{qr}^s are calculated.

- Magnitude λ_r and the phase θ_r of the rotor flux vector are determined using the rectangular to polar coordinate transformation.

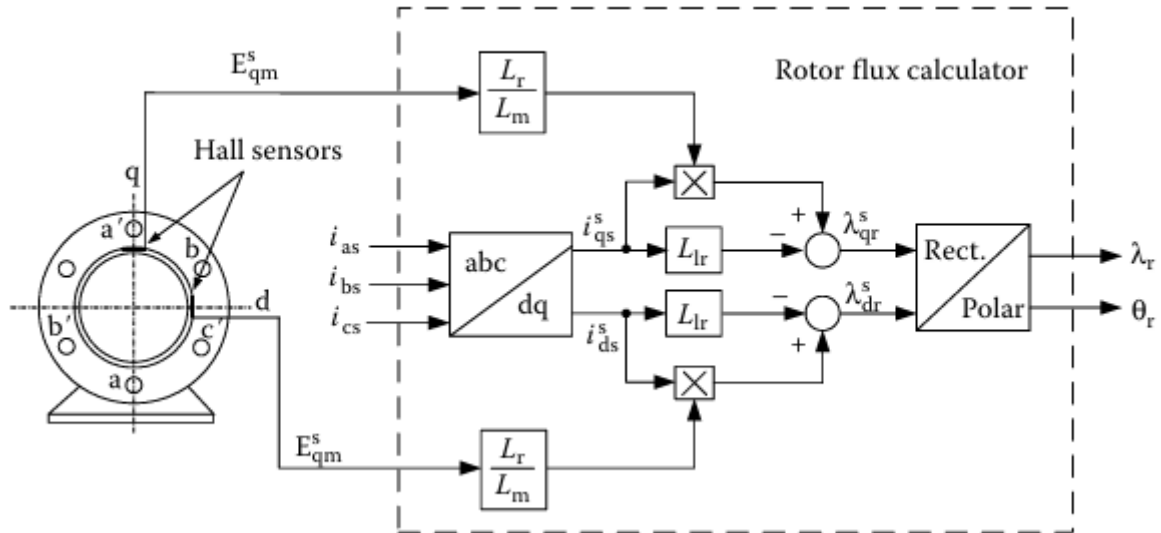


Figure 17: Determination of the magnitude and position of the rotor flux vector using a Hall sensor and a rotor flux calculator

It must be pointed out that the orthogonal spacing of the flux sensors in Figure 17 applies only to two-pole machines. In a P-pole machine, the sensors must be placed $180/P$ from each other.

Since $\lambda_{DR}^e = \lambda_r$ (see Figure 17), then the output variable, λ_r , of the rotor flux calculator can be used as a feedback signal in the field control loop. The same variables can also be used for calculation of the developed torque as shown in Figure 18.

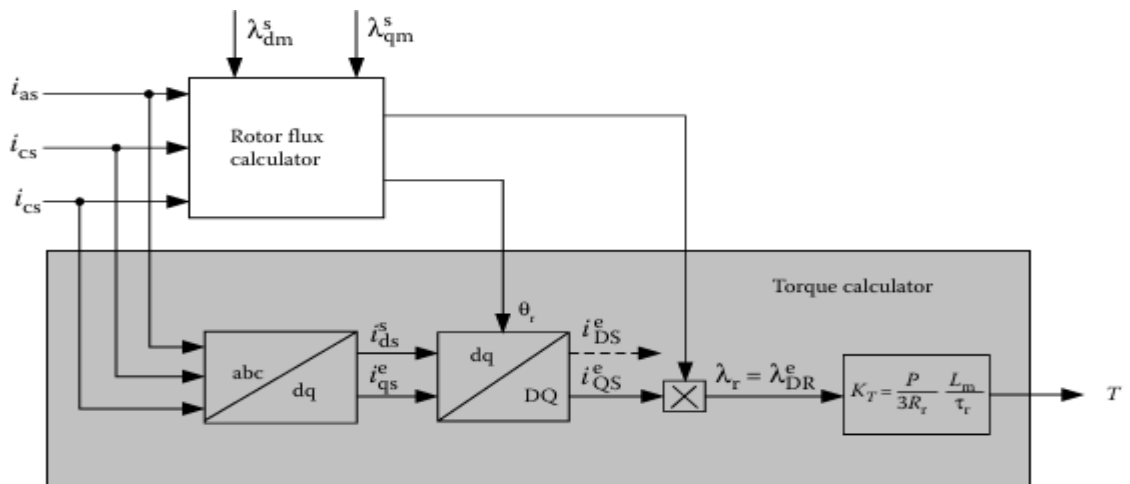


Figure 18: Torque Calculator

The torque calculator computes torque in the following steps:

1. The static abc to dq transformation is performed on the stator currents i_{as} , i_{bs} , and i_{cs} to obtain i_{ds}^s and i_{qs}^s
2. Angle θ_r supplied by the rotor flux calculator is substituted into Equation 6.63 for ωt in order to transfer signals i_{ds}^s and i_{qs}^s into i_{dS}^e and i_{qS}^e components of the stator current vector in the excitation frame.
3. Magnitude, λ_r , of the rotor flux, also supplied by the rotor flux calculator and presumed equal to λ_{dS}^e , is multiplied by i_{qS}^e and by the torque constant K_T to calculate the developed torque.

Figure 6.31 shows the torque calculation process block diagram.

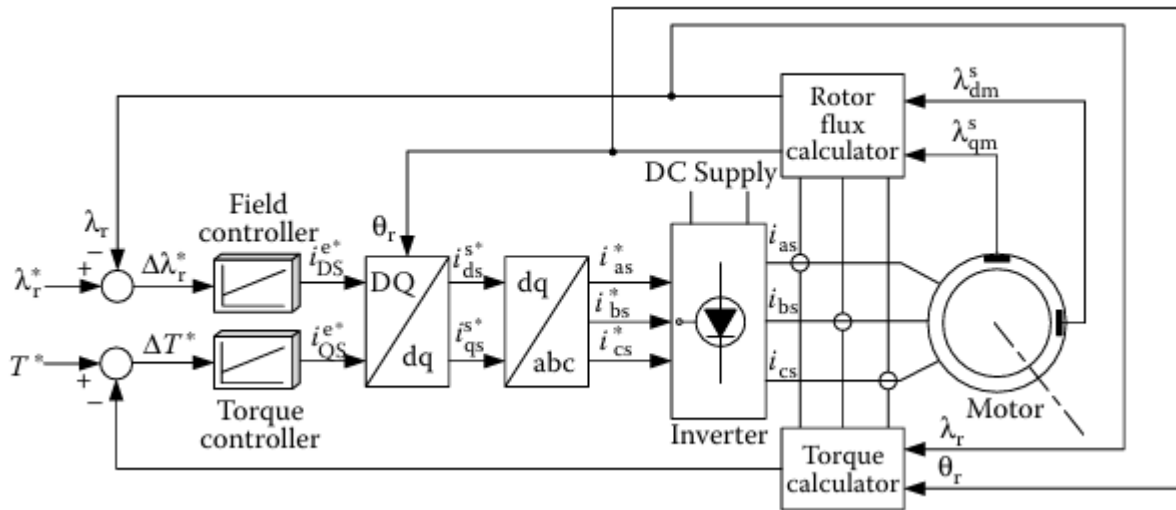


Figure 19: Vector control system for an induction motor with direct rotor flux orientation

Figure 19 illustrates an independent flux and torque control block diagram, based on the vector control of an induction motor with direct rotor flux orientation. In the system, proportional-plus-integral (PI)-based field and torque controllers are used to generate the control signals i_{dS}^{e*} and i_{qS}^{e*} in excitation frame by comparing the target rotor flux, λ_r^* , and target torque T^* , with the actual rotor flux, λ_r , and torque, T . Then, i_{dS}^{e*} and i_{qS}^{e*} in excitation frame are transferred into i_{ds}^{s*} and i_{qs}^{s*} in stator reference frame using rotor flux angle. Furthermore, i_{ds}^{s*} and i_{qs}^{s*} in stator reference frame are transferred into phase current signals i_{as}^* , i_{bs}^* , and i_{cs}^* through static transformation. The phase current signals, as the reference signals, are used to control the power electronics of the inverter to generate corresponding phase current i_{as} , i_{bs} , and i_{cs} .

In practice, the ratio of L_r to L_m , and the rotor leakage inductance, L_{ls} , which are used in the rotor flux calculator (see Figure 17), are not significantly affected by changes in the operation conditions of the motor, such as the winding temperature or saturation of the magnetic circuit. Therefore, the field orientation techniques described are considered to be the most robust and accurate. However, it requires the placement of vulnerable Hall sensors in the air gap of the motor, to the detriment of the cost and reliability of the drive system.

INDIRECT ROTOR FLUX ORIENTATION SCHEME

The presence of vulnerable Hall sensors in vector control with direct rotor flux orientation would weaken the reliability and enhance the cost of the motor drive. The indirect approach is to obtain the rotor flux position by the calculation of the slip speed, ω_r , required for correct field orientation, and the imposition of this speed on the motor.

If the synchronous speed necessary to maintain the orthogonal orientation of vectors λ_R^e and i_R^e in the given operating conditions of the motor is denoted by ω^* , the θ_r angle can be expressed as

$$\theta_r = \int_0^t \omega^* dt = \int_0^t \omega_r^* dt + \int_0^t \omega_0 dt = \int_0^t \omega_r^* dt + \theta_0$$

where ω^* , ω_r , and ω_0 are the synchronous speed, slip speed, and rotor speed, respectively, and θ_0 is the angular displacement of the rotor, which is easy to measure using a shaft position sensor.

The required value of the slip speed ω_r^* can be computed:

$$i_R^e = \frac{1}{L_r} (\lambda_{DR}^e - L_m i_S^e) \quad \text{Since } \lambda_R^e = \lambda_{DR}^e$$

Substituting above equation into Equation $R_r i_R^e + (p + j\omega_r) \lambda_R^e = 0$ gives the real and imaginary parts as

$$\lambda_{DR}^e (1 + \tau_r p) = L_m i_{DS}^e,$$

and

$$\omega_r \tau_r \lambda_{DR}^e = L_m i_{QS}^e$$

Replacing ω_r , λ_{DR}^e , and i_{QS}^e with ω_r^* , λ_r^* , and i_{QS}^* , respectively, in above Equation yields

$$\omega_r^* = \frac{L_m i_{QS}^e}{\tau_r \lambda_r^*}$$

Replacing λ_{Dr}^e and i_{DS}^e in Equation $\lambda_{Dr}^e(1 + \tau_r p) = L_m i_{DS}^e$, with λ_r^* and i_{DS}^{e*} yields

$$i_{DS}^{e*} = \frac{1 + \tau_r p}{L_m} \lambda_r^*$$

From the torque Equation, the signal i_{QS}^{e*} can be obtained as

$$i_{QS}^{e*} = \frac{T^*}{K_T \lambda_r^*}$$

A vector control system for an induction motor based on the indirect rotor flux orientation scheme is shown in Figure 20.

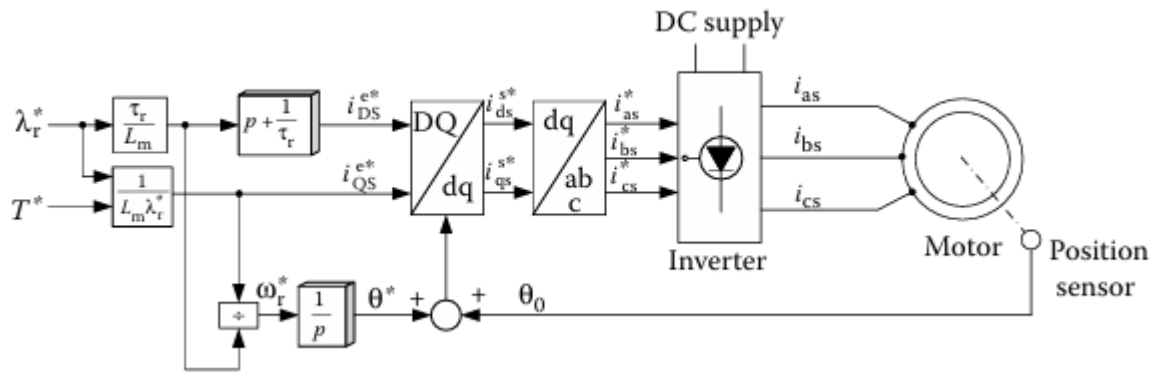


Figure 20: Vector control system for an induction motor with indirect rotor flux orientation

The rotor flux and developed torque are controlled in a feedforward manner. As a consequence of this, performance of the system strongly depends on an accurate knowledge of motor parameters, a requirement that is difficult to satisfy in practical applications. On the other hand, a major advantage of such a system is that a standard motor can be used, whose rotor position is easily measurable by an external sensor. Since the control scheme presented here constitutes an extension of the scalar torque control methods, the reference flux and torque values must satisfy the safe operation area condition described.